Horizon — the full pipeline (everything that was built)
green = math · lavender = LLM / spotlighting · rose = direct / jailbreak layer · white = code · beige = data / software

1 · Data (data/README.md = data card)

Indirect injection data
BIPIA (email · table · code, 15 attack kinds)
LLMail-Inject phase 2 (3,165)

AgentDojo v1.2.1
suites slack · workspace · travel
simulated tools, no real side effects

Corpus + canaries
attacks wrapped in carrier templates
CANARY-xxxxxxxx from secrets (CSPRNG)

Direct / jailbreak data
HackAPrompt (60,757 · 77,936 successful)
HarmBench · JBB · deepset · Gandalf
Tensor Trust · XSTest · Alpaca (benign)

Fixed splits
fit → choose τ*, π*, λ · calib · report
never report on the split used to choose

fit / report docs

2 · Stage 1 — keyed spotlighting (offline, chooses π*)

8 transforms τ
none · delimit · ^ datamark · N^ dynamic marker
ROT13 · shift (25) · scramble |Σ|! · Base64

Fresh key κ per prompt
keys.py → secrets

Spotlit prompt
$x_\tau(0) = S_\kappa \,\|\, \tau \,\|\, \tau_\kappa(d \oplus \delta)$

Attack indicator
$A(\delta, x) = \mathbb{1}[\text{canary appears in output}]$

Utility
U_τ = SQuAD token-F1 on clean docs

Clopper-Pearson
$\hat{p} \sim \text{Beta}^{-1}(1-\alpha;\ s+1,\ n-s)$

Forging bound
$q \,/\, |K_\tau|$ · (log-space)

Adaptive attacker (PAIR)
max over $\delta \in \Delta,\ |\delta| \le \text{attack_budget} \to \hat{p}^{\text{adp}}$

χ^2-DRO
$q_\rho(p) = p + \sqrt{(2\rho\, p(1-p))}$ (else 1)

Combined risk
$\hat{R}_\tau(\rho) = \min\{1,\ q/|K_\tau| + \max(q_\rho(\delta),\ \hat{p}^{\text{adp}})\}$

Sion game, level 1
$V_1 = \min_\pi \max_{\text{attack}}$ (SOCP, cvxpy/Clarabel) → π*

Choice rule
$\tau^* = \text{argmin}\ \hat{R}_\tau$ s.t. $U_\tau \ge U_{\text{none}} - \eta$

Significance
McNemar (paired) · bootstrap CI

Hand-off to Stage 2
π* · |K| = min |K_τ| · $\hat{R}_{\tau^*} \le \pi^*$, $\hat{R}_\tau$
results/stage1/pi_star.json

π*, |K|

RQ

3 · Direct injection / jailbreak layer (off-the-shelf, no training)

Output guard
Qwen3Guard-Gen 0.6B · Safe / Controversial / Unsafe

User prompt u
trusted channel · role owner / member / guest

Canary check
secret token leaked ⇒ block

Input guard
DeBERTa-v3 injection classifier (ProtectAI)

XPIA plants δ

Answer to the user

Role rules
policy.yaml: what each role may call

4 · Stage 2 — math-hardened CaMeL (online, every request)

Tool outputs
emails · files · web (untrusted)

Privileged LLM
sees ONLY u → writes a Python-like plan

CaMeL interpreter (STRICT)
google-research/camel, vendored unmodified
tracks data + control dependencies

untrusted x

Quarantined LLM × m
run i: own $\tau_i \sim \pi^*$, own fresh κ_i · no tools

Typed parser
Pydantic schema · strict / loose string caps

QIF leakage
$\Sigma \log_2|S|$ bits ≤ leakage_cap_bits

cap

Majority vote
> m/2 identical → value, else refuse

Agreement score
s(x) = (# agreeing runs) / m

Hoeffding
P(vote fooled) ≤ exp(−2m(½ − $\hat{R}_Q$)²)

Capability labels
SYSTEM ⊑ USER ⊑ UNTRUSTED
$\text{cap}(f(v_1,\ldots)) = \sqcup\ \text{cap}(v_i)$

Policy table
config/policy.yaml · strict / confirm_untrusted

Noninterference (Z3)
∀ args: UNTRUSTED ⇒ ¬ALLOW · proven at startup

Conformal risk control
ask iff s(x) ≥ λ, else DENY
$\hat{\lambda} = \min \lambda : (n/(n+1))\ \hat{R}(\lambda) + 1/(n+1) \le \alpha_{\text{user}}$

DENY

ALLOW

Ask the user
simulated approve / you in the shell

approved

Tool call
AgentDojo environment

G, C, F per attack

5 · End-to-end Horizon risk

Per-attack loss
$l = G + a{\cdot}C + h{\cdot}F$
G goal hit · C confirm asked · F false refusal

Loss bound
$\bar{l}$ = Clopper-Pearson upper on mean loss

Sion game, level 2 (LP)
$\min_{\text{cap}} \max_{\pi \in \Pi^*} \bar{l}$ · scipy linprog + duals ⇒ π*, w^*
Π* = configs with leakage ≤ cap and U ≥ $U_{\min}$

Horizon value
$V_H = \min\{1,\ \Sigma\pi\ q/|K| + \max|\text{EL}| + \sqrt{(2\rho\ \text{Var L})},\ \Sigma\pi\ \hat{p}^{\text{adp}}{}_\tau\}$
(variance form pending author confirmation)

0 · Software & infrastructure

uv + Python 3.11
pinned uv.lock · scripts/setup.sh

collibri engine
Qwen3.6-35B-A3B (t4)
writes the report

Math libraries
cvxpy + Clarabel (SOCP) · scipy (Beta⁻¹, linprog)
statsmodels · numpy · Z3

Ollama → Qwen3-8B
RTX 4060 · 16k ctx · q8 KV · flash attn
P-LLM, Q-LLM, attacker

MLflow
results/mlflow.db

Role proxy
developer→system · thinking off

LiteLLM
one client · on-disk cache

6 · Evaluation (§11 runners, logged to MLflow)

Unit tests (§9)
pytest · slide values only in tests/

Smoke test
does every layer fire?

CaMeL baseline
reference code on AgentDojo

stage1_runner
static · clean · adaptive · analyze

agentdojo_runner
ablations none · stage1 · stage2 · stage1+2 · full
calibrate λ · adaptive attacks

jailbreak_runner
detection rate / FPR on every direct set

Metrics
ASR = s/n with CP interval · utility · FPR

PyRIT
direct: PAIR · TAP · CrescendoJudge ≥ threshold)
indirect: XPIA plant δ → Horizon → canary

garak
direct: dan · promptinject · encoding
indirect: latentinjection · web_injection
bare model vs guarded_server.py

7 · Outputs & how you use it

./horizon.sh
menu · Stage 1 / Stage 2 / jailbreak demos
help · glossary · session logs

./run_all.sh
quick · pilot · full · --report-only

Figures F1–F18
matplotlib · PDF + PNG + CSV · formula box

Pipeline diagrams
Graphviz → docs/

Horizon.ipynb
every step, executed (nbclient)

report/report.pdf
colibri model analyses figures + CSVs
numbers checked → LaTeX (xelatex)